

Vancomycin exposure — a mini report (solution)

Your Name

today

1 Introduction

A single intravenous dose of a drug that follows first-order elimination is described by Equation 1, where D is the dose, V_d the volume of distribution and k_e the elimination rate constant.

$$C(t) = \frac{D}{V_d} e^{-k_e t} \quad (1)$$

2 Methods

Concentrations were read from `data/patient-concentrations.csv`. The elimination rate constant was estimated by least squares on the log-transformed concentrations, and the half-life followed from Equation 2.

$$t_{1/2} = \frac{\ln 2}{k_e} \quad (2)$$

```
conc <- read.csv("data/patient-concentrations.csv")

fit <- lm(log(conc_mgL) ~ time_h, data = conc)
c0 <- exp(unname(coef(fit)[1]))
k_el <- -unname(coef(fit)[2])
half_life <- log(2) / k_el
```

3 Results

Figure 1 shows the observed concentrations with the fitted curve. The fitted initial concentration is 19.9 mg/L and the estimated half-life is 3.1 h.

```
tt <- seq(0, 12, length.out = 200)

par(mar = c(4, 4.2, 1, 1), las = 1)
plot(conc$time_h, conc$conc_mgL,
     pch = 19, col = "#1b6ca8", cex = 1.1,
     xlab = "Time after dose (h)", ylab = "Concentration (mg/L)"
)
lines(tt, c0 * exp(-k_el * tt), col = "#b3502a", lwd = 2)
```

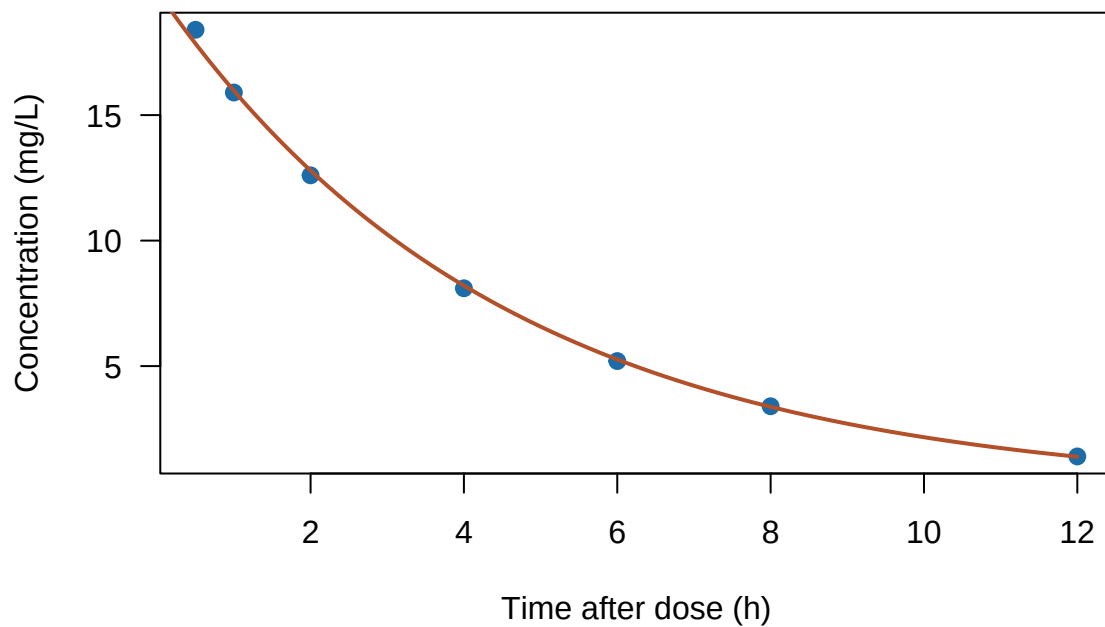


Figure 1: Observed concentrations and the fitted mono-exponential decay.

4 Summary

Table 1: Fitted parameters

Quantity	Estimate
C_0 (mg/L)	19.9
k_e (h ⁻¹)	0.222
$t_{1/2}$ (h)	3.1

The fitted values in Table 1 are computed at render time, so they cannot drift out of step with the figure in Figure 1.